

Submission STAT1-PS2-SUB01

Question STAT1-PS2-Q01

scratch: show step

1. $\bar{x} = 33/5 = 6.60$

2. $s^2 = \sum(x_i - \bar{x})^2 / (5-1) = 16.30$

=> FINAL: mean=6.60; sample variance=16.30 (check units/interpretation)

Question STAT1-PS2-Q02

scratch: show step

1. $P(D_1 \cap D_2) = (5/13)((5-1)/(13-1))$

2. =0.1282; the population shrinks after draw 1

=> FINAL: 0.1282 (check units/interpretation)

Question STAT1-PS2-Q03

scratch: show step

1. $SE = \sqrt{p(1-p)/n} = \sqrt{0.55(1-0.55)/40} = 0.0787$

2. $CI = 0.55 \pm 1.96(0.0787) = (0.396, 0.704)$

3. In repeated samples, the procedure captures the population proportion about 95% of the time.

=> FINAL: (0.396, 0.704); repeated-sampling interpretation (check units/interpretation)

Question STAT1-PS2-Q04

scratch: show step

1. Increase sample size n to reduce standard error.

2. Reduce confidence level or population variability when justified.

3. Lower confidence changes coverage, so precision is not free.

=> FINAL: increase n; lower confidence level or variability; note the coverage trade-off (check units/interpretation)

Question STAT1-PS2-Q05

scratch: show step

1. $0.03 < 0.05$, so reject H_0 under the stated rule.

2. Conclude evidence is inconsistent with H_0 ; do not say the probability H_0 is 3%.

=> FINAL: reject H_0 ; evidence against H_0 , not proof that H_0 is false (check units/interpretation)

Question STAT1-PS2-Q06

scratch: show step

1. $\bar{x} = 29/5 = 5.80$

2. $s^2 = \sum(x_i - \bar{x})^2 / (5-1) = 16.70$

=> FINAL: mean=5.80; sample variance=16.70 (check units/interpretation)

Question STAT1-PS2-Q07

scratch: show step

1. $P(D_1 \cap D_2) = (5/19)((5-1)/(19-1))$

2. ≈ 0.0585 ; the population shrinks after draw 1
=> FINAL: 0.0585 (check units/interpretation)

Question STAT1-PS2-Q08

scratch: show step

1. $SE = \sqrt{p(1-p)/n} = \sqrt{(0.45(1-0.45)/80)} = 0.0556$

2. $CI = 0.45 \pm 1.96(0.0556) = (0.341, 0.559)$

3. In repeated samples, the procedure captures the population proportion about 95% of the time.

=> FINAL: (0.341, 0.559); repeated-sampling interpretation (check units/interpretation)